

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460784.7255 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.02
Transit depth (Rp/Rs)^2	2.7 +/- 0.65 [%]
Orbital Inclination (inc)	81.02 +/- 1.76
Airmass coefficient 1 (a1)	0.956 +/- 0.011
Airmass coefficient 2 (a2)	0.03 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.19 %
Best Comparison Star	#1 - [486, 299]
Optimal Aperture	3.2
Optimal Annulus	7.03
Transit Duration (day)	0.0463 +/- 0.008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.164 ± 0.02 vs 0.1594 (deviation 0.23σ)
Duration fit vs published	0.0463 d vs 0.0512 d (deviation 0.62σ)
Mid-time O–C	1.3 min
Depth SNR	8.2
Residual scatter	1.19 %

Light curve

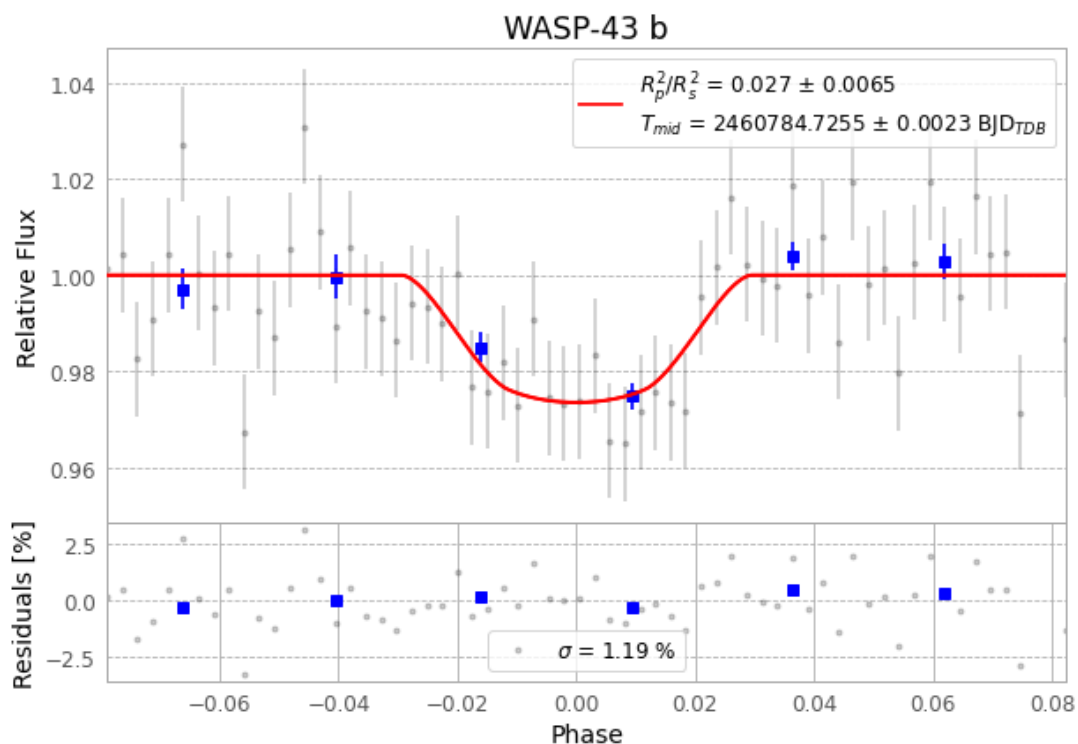


Figure 1 — FinalLightCurve_WASP-43 b_2025-04-18.png (SHA-256 f4914bdb3d89ff23...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [399, 252], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 907facd5c8fe29ba...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9e597cf

Data provenance

64 FITS frames (42 MB), WASP-43250419034515.FITS → WASP-43250419065416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250419.log (6cd4ead0c083...), FinalLightCurve_WASP-43 b_2025-04-18.png (f4914bdb3d89...), FinalLightCurve_WASP-43 b_2025-04-18.csv (cad74a6ce73c...)

Compute resources

Wall time 115.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 16:09Z.